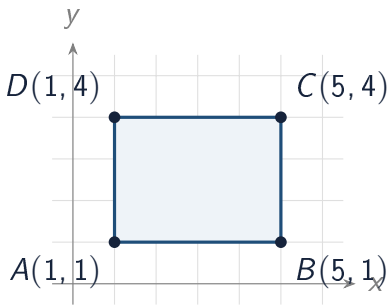
Lesson 1.4

Perimeter & Area in the Coordinate Plane

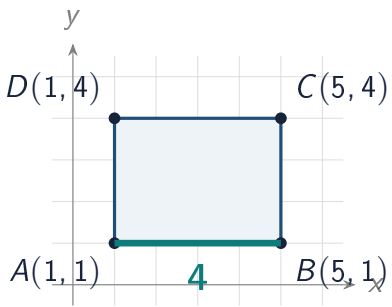
Count the sides, then use the formulas you know

Side lengths from coordinates



Rectangle $ABCD$

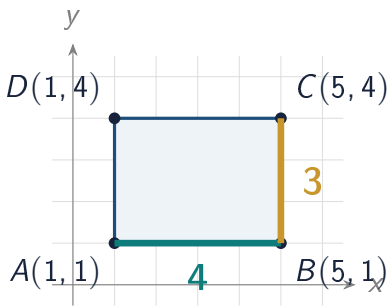
Side lengths from coordinates



Rectangle $ABCD$

$AB = 5 - 1 = 4$

Side lengths from coordinates

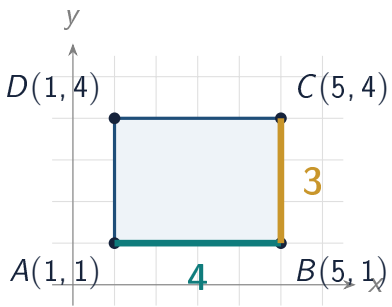


Rectangle $ABCD$

$AB = 5 - 1 = 4$

$BC = 4 - 1 = 3$

Side lengths from coordinates



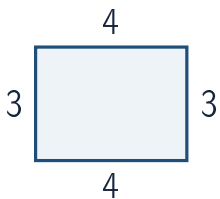
Rectangle $ABCD$

$AB = 5 - 1 = 4$

$BC = 4 - 1 = 3$

Same y: subtract the x's.
Same x: subtract the y's.

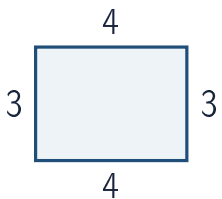
Perimeter and area



Perimeter = add all the sides

$P = 4 + 3 + 4 + 3 = 14$

Perimeter and area



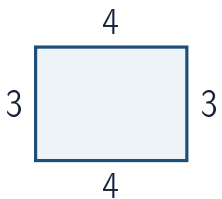
Perimeter = add all the sides

$P = 4 + 3 + 4 + 3 = 14$

Area = base $\times$ height

$A = 4 \cdot 3 = 12$

Perimeter and area



Perimeter = add all the sides

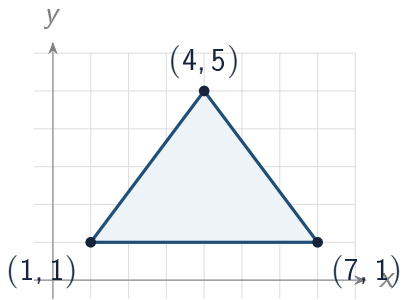
$P = 4 + 3 + 4 + 3 = 14$

Area = base $\times$ height

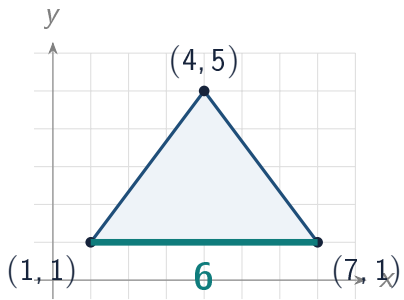
$A = 4 \cdot 3 = 12$

Perimeter is a length: **units**. Area counts squares: **square units**.

Area of a triangle

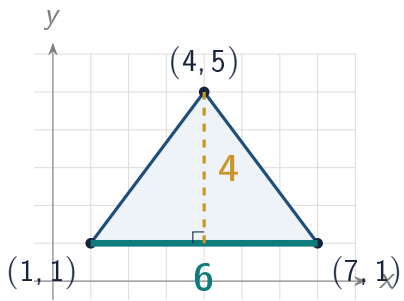


Area of a triangle



$\text{base} = 7 - 1 = 6$

Area of a triangle

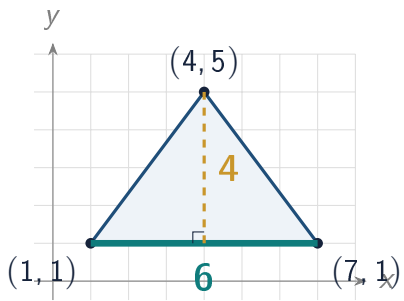


base $= 7 - 1 = 6$

height $= 5 - 1 = 4$

straight up from the base

Area of a triangle



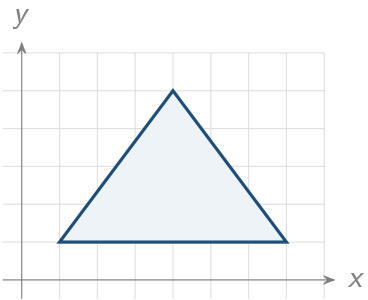
base $= 7 - 1 = 6$

height $= 5 - 1 = 4$

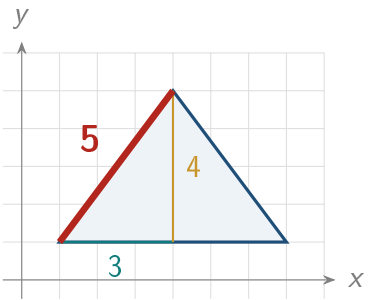
straight up from the base

$$A = \frac{1}{2} \cdot b \cdot h = \frac{1}{2} \cdot 6 \cdot 4 = 12$$

A slanted side? Use Pythagoras



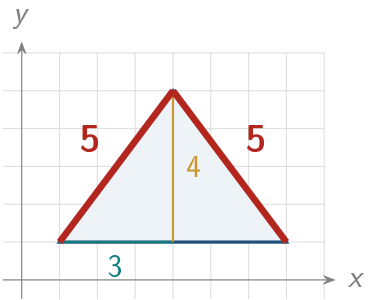
A slanted side? Use Pythagoras



Across 3, up 4:

$$\sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

A slanted side? Use Pythagoras



Across 3, up 4:
$\sqrt{3^2 + 4^2} = \sqrt{25} = 5$

The other side is also 5.

$P = 6 + 5 + 5 = 16$

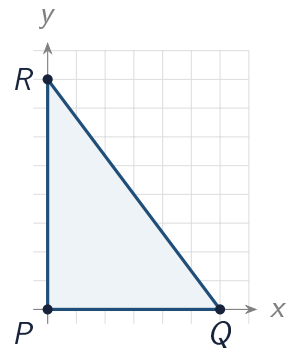
Watch out

Watch out! For a **slanted** side you can't just count squares. Use the Distance Formula — or draw the right triangle and use Pythagoras.

Your turn! Pause the video and try it.

Triangle $P(0,0)$, $Q(6,0)$, $R(0,8)$.

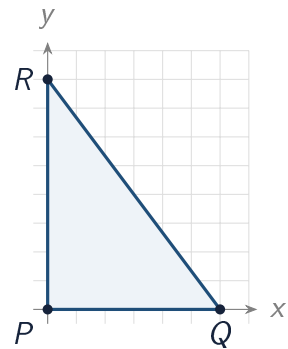
Find its perimeter and area.



Your turn! Pause the video and try it.

Triangle $P(0,0)$, $Q(6,0)$, $R(0,8)$.

Find its perimeter and area.



$PQ = 6,\ PR = 8$

$QR = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$

$P = 6 + 8 + 10 = 24$

$A = \frac{1}{2} \cdot 6 \cdot 8 = 24$

Remember

- Same y: subtract the x's. Same x: subtract the y's.
- **Perimeter**: add the sides. **Area**: rectangle $b \cdot h$, triangle $\frac{1}{2}bh$.
- Slanted side: Pythagoras or the Distance Formula.

Now practice at your level:

fangmath.com → Geometry → Chapter 1